

NCEA Numeracy: The "Justify Your Thinking" Worksheet

The Strategy: The "I.C.E." Method

To achieve, your written answers should follow this flow:

1. **Information:** What numbers/units are you using?
2. **Calculation:** What is the math result?
3. **Explanation:** What does that number mean in the real world?

Task 1: The "Best Buy" (Full Scaffolding)

The Problem: Store A sells a skateboard for \$120. Store B sells the same board for \$150 but has a "20% Off" sale. Which store is cheaper?

Step 1: The Math

- Store A Price: \$120
- Store B Discount: $0.20 \times \$150 = \$\$$ _____
- Store B Final Price: $\$150 - \$\$$ _____ $= \$\$$ _____

Step 2: The Written Response "I believe Store _____ is the better deal. Even though Store B has a sale, their final price is \$_____, while Store A is only \$_____. This means Store _____ saves the customer \$_____."

Task 2: The Trip Planner (Partial Scaffolding)

The Problem: You are organizing a trip to a climbing gym. There are 26 students, and each van can hold 7 people. How many vans must you book?

Step 1: The Math

- Calculation: _____
- Calculator Result: _____

Step 2: The Written Response

"I need to book _____ vans. I chose this number because..."

Task 3: The Data Detective (Low Scaffolding)

The Problem: A graph shows the "Mean" (average) temperature in Dunedin is 10°C , but today it is 14°C . A student claims today is "unusually hot" compared to the average.

Evaluate this claim.

Step 1: The Math (Show your working for the difference or comparison here)

Step 2: The Written Response (State whether you agree or disagree with the claim and explain **why** using your math)

Task 4: The Independent Navigator (No Scaffolding)

The Problem: A map has a scale where $1\text{cm} = 2\text{km}$. You need to deliver a package to a house that is 7.5cm away on the map. You have enough fuel to travel 14km . Do you have enough fuel for the trip?

Your Response: (Use the space below to show your calculation and write your final conclusion. Remember to check if your answer is appropriate for the context) .



Pro-Tips for Tuesday:

- **Units are Mandatory:** Always include \$, kg, m, or $\text{^\circ C}$.
- **Follow Exact Instructions:** If it says "show your working," don't leave the box blank.
- **The "Because" Rule:** Your explanation is only complete if you explain **why** the math matters to the person in the story.